

CHE 205, Fall 2019, Problem Set 1

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Due date: 6:00pm, Fri, 09/06/2019

All problems are from the textbook by V.L. Law.
Submit the spreadsheet on blackboard by the due date.

Problem 1 (7 pts). Exercise 1.1.

Problem 2 (7 pts). Exercise 1.2.

Problem 3 (7 pts). Exercise 1.3.

Problem 4 (7 pts). Exercise 1.6.

Exercise 1.1: For the following functions, graph the function and then use Goal Seek to find the root(s).

a. $f(x) = x - x^{1/3} - 2$

b. $f(x) = x \tan x - 1$

c. $f(x) = x^4 - e^x + 1$

d. $f(x) = x^2 e^x - 1$

Exercise 1.2: Find the roots of the functions given in Exercise 1.1 using the bisection method. Use the graph of each function to choose points that bracket the root of interest.

Exercise 1.3: Set up a spreadsheet that implements the secant method and then solve each of the problems from Exercise 1.1. Use the graph of each function to select an initial guess. Recall the iteration formula for the secant method:

$$x^{k+1} = x^k - \frac{f(x^k)}{f(x^k) - f(x^{k-1})}(x^k - x^{k-1})$$

Hint: Set up the first row of the spreadsheet for your problem with headings such as the following:

x_{k-1}	x_k	$f(x_{k-1})$	$f(x_k)$	x_{k+1}
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Put the formula for the function under the headings $f(x_{k-1})$ and $f(x_k)$. In the cell under x_{k+1} , put the secant method iteration formula. In the second row, replace the previous x_{k-1} with x_k and then x_k with x_{k+1} . Now copy the two formulas down one row. At this point, one iteration of the secant method is displayed. To see more iterations, just copy the second row down for as many iterations as desired. If too many iterations are copied and the function difference (the denominator of the iteration formula) becomes exactly zero, a “divide by zero” error will appear.

Exercise 1.6: Solve the problems of Exercise 1.1 using the Newton method.